

THE AQUACULTURE PROFIT LEAK AUDIT

7 HIDDEN COSTS KILLING SMALL OPERATORS
AND THE 30-MINUTE FIX FOR EACH ONE



ENG. HAZEM SHANNAK

SHRIMP OPERATOR FIELD GUIDE SERIES — VOL. 01

THE AQUACULTURE PROFIT LEAK AUDIT

*7 Hidden Costs Killing Small Operators
And the 30-Minute Fix for Each One*

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OPERATOR-GRADE FIELD GUIDE

Before You Turn The Page

I have written this book three times.

The first version was polite. It had charts and case studies and careful language. I deleted it.

The second version was angry. It blamed everyone – hatcheries, feed companies, the weather. I deleted that too.

This is the third version.

It is not polite and it is not angry, but It is useful.

Here is what I promise you:

By the time you finish this book, you will know exactly where your operation is leaking money. You will not need a consultant to tell you. You will not need a spreadsheet model. You will need a pen, your harvest records, and a 30-minute walk around your ponds.

Here is what I ask in return:

Do not skip around. Read the leaks in order. Score yourself honestly. And when you see a 2 or a 3 on that scorecard, do not close the book and say "I'll deal with that later."

That is the deal. If you do your part, I guarantee you will find at least one leak that pays for this book a hundred times over before your next harvest.

Now turn the page and let's go to work.

First Note Before We Begin

If you run a fish or shrimp operation doing revenue somewhere between \$15K and \$100K a month, this short book is written for you. Not for the textbook crowd. Not for the YouTube influencers selling courses on species they have never grown but for real farm operators.

I have spent thirty years rebuilding under performing aquaculture ventures into investment-ready businesses across three continents. In that time I have seen the same profit leaks drain margins from small operators over and over again.

They are not the leaks the consultants talk about.

They are the quiet ones.

This audit is free because it is the diagnostic. The fix work is paid.

But before you pay anyone - including me - you need to know where you are actually bleeding.

Most operators I meet are guessing.

After this read, you will not be.

Read it once. Then walk your farm with it open on your phone. Score yourself honestly.

The whole thing is designed to take you under two hours, including the walk.

Hazem

How To Use This Audit

Each of the 7 leaks below has the same structure:

1.	THE LEAK	What it actually is, in plain operator language.
2.	WHY IT'S HIDDEN	Why most operators don't see it until it's expensive.
3.	30-MIN DIAGNOSTIC	What to check, today, on your operation.
4.	THE FIX	The lowest-cost first move.
5.	REAL COST IMPACT	What this leak typically costs in % margin.

At the end you will find a one-page scorecard. Print it. Score each leak from 0 (not a problem) to 3 (bleeding badly). Anything scoring 2 or 3 needs work this quarter.

THE PRE-AUDIT WARM-UP

15 Minutes. A Pen. Your Last Three Batch Records.

Before you walk your farm, you need three numbers. Do not skip this. Operators who skip this spend the whole audit guessing. Guessing does not fix leaks.

NUMBER 1: Your average harvest kg per cycle across the last three cycles.

Write it here: _____

NUMBER 2: Your average feed cost per kg of harvest across the last three cycles.

(Total feed spent ÷ total kg harvested)

Write it here: _____ per kg

NUMBER 3: Your average survival rate across the last three cycles.

(Final head count ÷ stocked head count) × 100

Write it here: _____

If you cannot find these numbers in 15 minutes, stop. Your first leak is not in the seven below. Your first leak is that you are not tracking batch performance at all. Go set up a simple spreadsheet. One row per pond per batch. Date stocked, number stocked, date harvested, number harvested, kg harvested, total kg feed used. That is it. Come back when you have three rows.

If you have the numbers but they hurt to look at, good. That is the point.

1

LEAK

THE FEED CONVERSION DRIFT



Every point of FCR drift costs thousands per cycle — and most operators never notice it happening.

LEAK NO. 01

The Feed Conversion Drift

The Feed Inventory Lie & The Shrimp-Specific FCR Trap

THE LEAK

Feed is 50–70% of your variable cost. Your Feed Conversion Ratio (FCR) is the single biggest lever you have. Most small operators calculate FCR once when they start a batch, then never recalculate as conditions change.

There are water temperature shifts, pellet quality changes between batches, and fish age changes their conversion efficiency. Your FCR drifts up and you don't notice, because you are still feeding by the table on the bag.

With shrimp, FCR is trickier than fish. Shrimp feed on the bottom. They take longer to eat, they are shredders not gulpers. A tilapia eats a pellet whole, but shrimp picks at it, breaks it apart, and leaves fines that become wasted protein that turns into ammonia.

Most shrimp operators calculate FCR on total feed dropped, not feed consumed, and that difference is often 10–15% right there. The feed bag says 1.2, but your real FCR including waste and fines is 1.4.

You never knew because you never swept the pond bottom to see what was left.

WHY IT'S HIDDEN

The drift is gradual. A tilapia operation running at 1.6 FCR can drift to 1.9 over six months without anyone flagging it. You don't see it because you are measuring inputs (kg of feed) and outputs (kg of fish at harvest) on different timelines. By the time the harvest reveals the bad FCR, the batch is gone.

Three reasons specific to shrimp:

1- No stomach sample. With fish, you can gut-sample and see fill. With shrimp, you need a scope to check gut fullness in the hepatopancreas. Almost nobody does this weekly.

2- Bottom waste looks like sediment. Uneaten feed breaks down into a sludge that looks like natural pond muck. You scrape it, shrug, and call it normal.

3- Shrimp eat their own waste. They will re-ingest fecal matter and broken feed pellets. That makes it look like they ate the feed when they actually just recycled waste. Your FCR looks better on paper than it is in reality.

THE 30-MINUTE DIAGNOSTIC

Pull your last three harvest records. Calculate actual FCR per batch:

FCR = Total kg of feed used ÷ Total kg of biomass gained

If you cannot do this calculation in under 30 minutes, that is your first leak — you cannot fix what you do not measure.

Compare your actual FCR to the species benchmark:

- Tilapia: 1.4–1.7 is good
- Catfish: 1.5–1.8
- Salmon (RAS): 1.0–1.3
- Shrimp (vannamei): 1.2–1.6

Anywhere above the high end means money is leaving every day.

After you calculate your batch FCR, run this second calculation:

Estimated waste feed = Total feed fed × 0.15 (assumed waste for shrimp)

Then subtract that from total feed before recalculating FCR. If your adjusted FCR jumps more than 0.2 points higher than bag claim, your feeding method is leaking.

Then do this: pick one pond. Feed it normally. One hour later, take a bottom sampler (a core tube or a simple PVC pipe with a cap). Pull three samples from the feeding zone. Spread the sludge on a white tray. Count how many whole pellets and pellet fragments you see. If you see more than 5–10 pellets per sample, you are overfeeding.

THE FIX

Start weekly biomass sampling, even rough estimates beat no estimates. Adjust feed rate to actual biomass, not bag tables. Cut hand-feeding waste by switching to demand feeders or fixed feeding rings on at least your largest ponds.

Install feed trays. Cheap and simple, a 60cm round tray, 2cm mesh bottom, suspended just above the pond bottom. Put one tray per 500–1000 square meters of pond. Drop your feed. After 45–60 minutes, pull the tray. If feed remains, you are done feeding for that meal. If the tray is clean, you can consider adding more.

One farmer I worked with in Indonesia cut his shrimp FCR from 1.7 to 1.4 in one cycle using nothing but feed trays and a rule: no feed left on the tray after one hour, ever.

REAL COST IMPACT

- Going from 1.9 FCR to 1.6 FCR on a 10-tonne tilapia operation at \$1.20/kg feed = **\$3,600 saved per cycle**. Three cycles a year = over \$10K. That's one leak.

- A shrimp operation doing 10 tonnes per cycle at 1.7 FCR vs 1.4 FCR on feed at \$1.10/kg:

- 1.7 FCR = 17,000 kg feed = \$18,700
- 1.4 FCR = 14,000 kg feed = \$15,400
- Savings = \$3,300 per cycle

Three cycles a year = **\$9,900**. From feed trays costing less than \$20 each.

My score for this leak: ____ / 3

LEAK 2

INVISIBLE MORTALITY



*Between 8% and 25% of your stock vanishes
before harvest with no body and no
explanation.*

LEAK NO. 02

Invisible Mortality

The Molt Mortality & The Cannibalism That You Don't See

THE LEAK

The fish that die in the open and float, those you count. The fish that die at the bottom, or get cannibalized, or die slowly from chronic poor water quality, those become a 'stocking error' or a 'harvest discrepancy.' Most small operators have invisible mortality rates between 8% and 25% that they are not tracking.

Shrimp die differently than fish. Fish float when they die (most of them). Shrimp sink. Then they get eaten by live shrimp within hours. By the time you look for dead shrimp, there is nothing to find. No body, no log entry, no problem in your spreadsheet. Just a harvest that came up light.

But here is the part almost nobody talks about: molt mortality. Shrimp shed their shell to grow and during molt, they are soft, immobile, and cannot escape predators including other shrimp. In high-density ponds, mortality during molt can hit 5–10% and you will never see a single dead body. The live shrimp eat the molting shrimp before they harden.

WHY IT'S HIDDEN

There is no dead body. The biomass just isn't there at harvest. Operators round it up to 'shrinkage' or blame the fingerlings. Hatcheries get blamed for problems that started in your water column.

You log dead shrimp you see floating or at the edge. Those are the strong sick ones that came to the surface, yet the weak ones die on the bottom and disappear. Your mortality log might show 3% loss. Your actual survival might be 75%. That 22% gap is invisible.

Also: cannibalism looks like slow growth. A shrimp that eats a molting neighbor gains protein and it grows a bit faster. You see some shrimp looking good and think the pond is fine. Meanwhile your overall population is crashing.

THE 30-MINUTE DIAGNOSTIC

Compare your stocking number to your harvest number for the last three batches. The gap is your total mortality. Subtract any documented mortality (the floats you logged). What's left is invisible mortality.

If invisible mortality is more than 5%, you have a problem. If it is more than 15%, your operation is bleeding heavily and you don't know where.

You need two numbers you probably do not have:

1. **Post-molt count.** At night (shrimp molt mostly at night), shine a light in shallow water. Count soft shrimp clinging to the edge or hiding. Do this same time each week. If post-molt shrimp are more than 2–3% of the visible population, you have a molt mortality problem.

2. **Cannibalism check.** Toss a small seine in the pond. Pull 50–100 shrimp. Look for fresh bite marks on the tail fans and legs. More than 10% of shrimp showing healed or fresh bite marks means density is too high or feed is insufficient.

THE FIX

Three changes:

1. Daily visible mortality logs — yes, every day. Number, size, suspected cause.
2. Bottom checks on at least one pond per week using a viewing scope or a quick seine.
3. Water quality logs at the same hours daily, not whenever someone remembers.

For molt mortality: add sheltered substrate. Simple horizontal netting, mesh bags, or old feed bags hung vertically in the water. Gives molting shrimp a place to hide during their soft window. In one Thai operation, adding 10% substrate surface area relative to pond area cut invisible mortality from 22% to 11% in one cycle.

For cannibalism: increase feeding frequency, not total feed. Same daily ration, split into 4–5 meals instead of 2–3. Keeps feed in the water more consistently, reduces hungry shrimp looking for soft neighbors.

REAL COST IMPACT

On a tilapia farm stocking 50,000 fingerlings, dropping invisible mortality from 18% to 8% recovers 5,000 fish. At 600g harvest weight and \$2.20/kg = **\$6,600 recovered per cycle.**

That is not a leak. That is a hole in the floor.

My score for this leak: ____ / 3

LEAK **3**

THE STOCKING DENSITY MISFIT



*Running at the wrong biomass. Both
overcrowding and underuse drain margin in
opposite directions.*

LEAK NO. 03

The Stocking Density Misfit

The Oxygen Demand Per Kg – Shrimp Are Not Fish

THE LEAK

You stocked at the density a textbook or another farmer told you to use. But your water exchange rate, your aeration capacity, your average water temperature, and your management hours are not the same as theirs. So you are either under-stocking (leaving capacity on the table) or overstocking (stunting growth and raising mortality).

Shrimp consume less oxygen per kg than fish. That sounds good. But they are also less tolerant of low oxygen. Fish can survive down to 3–4 mg/L for short periods. Shrimp start dying at or below 3 mg/L and stop feeding well below 4.0 mg/L.

So your real carrying capacity is not set by the oxygen they need to survive. It is set by the oxygen they need to feed and grow. That threshold is higher.

WHY IT'S HIDDEN

Both versions of this leak look like normal operation. Under-stocking just looks like 'we had a quiet cycle.' Overstocking looks like 'the fish grew slow this time.'

Almost every textbook oxygen consumption rate for shrimp is measured in lab conditions with clean water, no stress, and shrimp that are not buried in the mud. In a real pond, at peak biomass, with a sediment layer consuming oxygen, your actual available oxygen for shrimp is 40–60% of what your aerator produces on paper.

The other hidden piece: heterogeneous distribution. Fish fill the water column. Shrimp stay on the bottom. The bottom 20cm of water has lower DO than the surface, even with aerators running. You measure DO at 50-60 cm depth and think you are fine. The shrimp at the bottom are suffocating.

THE 30-MINUTE DIAGNOSTIC

Calculate your actual carrying capacity — not the one on a forum post. Multiply your dissolved oxygen production capacity (your aerators' rated kg-O₂/hour, derated 30% for real-world conditions) by 24 hours. Divide by the species' oxygen consumption rate at your average temperature. That is your real biomass ceiling, not the one in a brochure.

For tilapia at 28°C, fish consume about 350 mg-O₂/kg/hour. So an aerator producing 1 kg-O₂/hour (real-world) supports about 2,850 kg of standing biomass — at the maximum, before any safety margin.

Measure DO at the bottom. Not mid-water. Not at the side. Use a weighted DO probe or lower a bucket fast, pull it up, and measure immediately. Do this at 4am (lowest DO of the day) and at 2pm (peak feed time).

If bottom DO is more than 1.5 mg/L lower than surface DO, your aeration is not reaching the bottom. Your true carrying capacity is 30–50% lower than your calculation.

THE FIX

Re-calculate every pond's carrying capacity using your actual aerator output, not the rated output. Apply a 30% safety margin. If you are below capacity, increase stocking on the next cycle. If you are above, you have found your bottleneck.

First: redirect aerators. Most operators point aerators to make a circular flow. That creates a vortex that pulls water off the bottom but does not mix it. Instead, angle aerators to create turbulent mixing that reaches the bottom. You want bubble plumes hitting the pond floor, not just surface rotation.

Second: bottom aeration. One simple diffuser per pond, even a single line of perforated pipe weighted to the bottom, run off a small blower. This single addition typically raises bottom DO by 1–2 mg/L and allows 20–30% higher stocking density with the same survival.

REAL COST IMPACT

Most operators I audit are running 20–40% below carrying capacity in their best ponds. On a \$40K/month operation, that is **\$8K–\$16K of capacity unused every month.**

A shrimp farm running 50 ponds at 80% of true carrying capacity (because they are oxygen-limited at the bottom). Increasing stocking density 25% on the next cycle:

- Current harvest: 40 tonnes per cycle
- New harvest: 50 tonnes per cycle
- At \$5/kg = \$50,000 additional revenue per cycle

Cost to add bottom aeration to 50 ponds: roughly \$8,000–12,000 one time. Paid back in the first partial cycle.

My score for this leak: ____ / 3

LEAK 4

THE WATER EXCHANGE BLACK HOLE



*Fixed exchange rates pump money out with
the water, regardless of what the pond
actually needs.*

LEAK NO. 04

The Water Exchange Black Hole

THE LEAK

Either you are exchanging too much water (paying for pumping energy and losing nutrients/heat) or too little (toxin buildup eats into FCR and immunity). Most operators pick a number — say 10% per day — and stick to it regardless of biomass, temperature, or feed load.

Shrimp are more sensitive to water chemistry swings than fish. A rapid change in salinity or alkalinity triggers molting. Molting triggers cannibalism. Cannibalism triggers invisible mortality. You changed 20% of your water yesterday and now shrimp are dying and you have no idea why.

The other hidden exchange cost: phytoplankton crash. When you exchange water, you remove algae. When you remove algae, ammonia spikes. When ammonia spikes, you exchange more water. It is a death spiral that costs you in power, water, and shrimp health.

WHY IT'S HIDDEN

Both extremes look like normal cost-of-doing-business until they don't. Over-pumping is a steady electricity bill that nobody questions. Under-exchange shows up as 'the fish aren't eating well this week' — masked by a hundred other variables.

Most shrimp operators track salinity at exchange time. They do not track alkalinity. Alkalinity buffers pH. When alkalinity drops below 80 ppm, pH swings start. When pH swings more than 0.5 points in 24 hours, shrimp get stressed. Stressed shrimp stop feeding. Stopped feeding means wasted feed. Wasted feed means ammonia.

The whole chain starts with an alkalinity number you are probably not measuring.

THE 30-MINUTE DIAGNOSTIC

Test ammonia (TAN) and nitrite at peak feed time (usually mid-afternoon). If TAN is over 1.0 mg/L unionized or nitrite is over 0.5 mg/L, you are under-exchanging. If both are consistently near zero and your electricity bill is more than 12% of revenue, you are over-exchanging.

Test alkalinity before and after every water exchange for one week. If post-exchange alkalinity is more than 20% lower than pre-exchange, your source water is low-alkalinity and every exchange is making your system less stable.

Test the same for salinity. If source water salinity differs from pond salinity by more than 5 ppt, each exchange is forcing a molt cycle. That costs you feed conversion and survival.

THE FIX

Move from a fixed exchange rate to a trigger-based exchange rate. Set thresholds. Exchange when TAN or nitrite hit them, not on a calendar.

Stop exchanging water to manage water quality. Exchange water to manage ionic balance only. Use biofloc or probiotics to handle ammonia and nitrite without dumping water.

If you must exchange, do it slowly. A continuous trickle exchange (1–2% per hour over 10 hours) causes less shock than a single 20% dump and fill. Shrimp tolerate gradual change. They do not tolerate sudden change.

REAL COST IMPACT

On a flow-through operation, cutting unnecessary exchange by 40% can drop pumping costs by **\$400–\$1,200 per month** depending on scale. On a static system with under-exchange, fixing it can recover 0.2–0.4 points of FCR (see Leak #1 for what that's worth).

A brackish water shrimp farm exchanging 15% daily at 2kW pumping cost per pond. Switching to 5% daily continuous exchange:

Pumping cost reduction: 66% less energy

On 20 ponds at \$200/month pumping each = \$4,000/month current

New cost = \$1,350/month

Savings = \$2,650/month = \$31,800/year

And that is before you count the feed conversion improvement from stable water chemistry.

My score for this leak: ____ / 3

LEAK 5

ENERGY: THE SILENT MARGIN KILLER



Aerators running for ghosts. Equipment sized for peak conditions, burning kWh 24 hours a day.

LEAK NO. 05

Energy: The Silent Margin Killer

THE LEAK

Aerators and pumps were sized for peak conditions. They run at full power 24/7 even when biomass is half of capacity. Most operators have never measured kWh per kg of fish produced.

Shrimp ponds need more aeration per kg of biomass than fish ponds. Shrimp are bottom dwellers, as we covered. Bottom water has lower DO. Lower DO means more aerators. More aerators mean more electricity.

But here is the part nobody checks: aeration efficiency drops 30–50% within six months if you never clean your aerators. Fouling, biofouling, and hard calcium deposits reduce oxygen transfer. You think you are putting 2 kg-O₂/hour into the water. You are actually putting 1.2. Your carrying capacity is not what you calculated. It never was.

WHY IT'S HIDDEN

The bill comes monthly. It is a fixed-feeling number. Nobody breaks it down per kg of harvest, so nobody knows whether it is reasonable.

You measure DO at the surface or mid-water. You see acceptable numbers. You assume your aerators are working. But the oxygen is not reaching the bottom because your aerators are producing big bubbles (inefficient transfer) and the bubbles are rising straight up (poor mixing).

The other hidden piece: time of day. Most operators run aerators 24/7 after the first month. At night, when photosynthesis stops, DO drops. That is when you need aeration. During peak daylight, when algae are producing oxygen, you often do not need full aeration. But you run them anyway.

THE 30-MINUTE DIAGNOSTIC

Take your last 12 months of electricity bills. Take your last 12 months of fish harvested in kg. Divide. Benchmark:

- Pond systems: 0.4–0.8 kWh per kg fish
- Intensive flow-through: 1.5–2.5 kWh per kg
- RAS: 4–8 kWh per kg

If you are meaningfully above these ranges, your energy is leaking somewhere — likely an oversized pump or aerators running when DO is already adequate.

Walk your pond bank at 2pm on a sunny day. Turn off aerators on one pond for 30 minutes. Measure DO at the bottom at the start and at the 30-minute mark. If DO does not drop below 5 mg/L, you are over-aerating during daylight hours on that pond.

Then pull one aerator from each pond type. Clean it thoroughly. Run it in a barrel of clean water with a DO meter. Measure oxygen transfer rate. Compare to manufacturer spec. If you are below 70% of spec, every aerator on your farm is under-performing.

THE FIX

Install one DO sensor with a relay (under \$400) on your largest pond. Set aerators to cycle off when DO is above 6 mg/L. This single change typically cuts aerator runtime 25–40%.

First: clean schedule. Aerators pulled and cleaned every 60–90 days. Not when you remember. Not when the motor sounds funny. Every 60 days, on the calendar.

Second: timer banks. Cheap mechanical timers (under \$50 each). Set aerators to run on a cycle: 30 minutes on, 30 minutes off during daylight hours when DO is high. Full aeration only at night and during cloudy weather. This single change cut power use 35% on a farm I worked with in Vietnam without any measurable impact on survival or growth.

REAL COST IMPACT

On an operation using 15,000 kWh/month at \$0.12/kWh, a 30% aeration reduction saves about **\$540/month — \$6,500/year**.

A 20-pond shrimp farm running 8 aerators per pond, 1.5 kW each. 24 hours a day. Power at \$0.14/kWh.

- Current monthly aeration power = 20 ponds × 8 aerators × 1.5 kW × 24h × 30 days = 172,800 kWh
- Cost = \$24,192/month

Switch to 12 hours of aeration per day (nighttime only + daytime cycling). New consumption = 86,400 kWh. New cost = \$12,096/month. Savings = \$12,096/month = \$145,000/year.

You could hire two full-time farm managers for that money.

My score for this leak: ____ / 3

LEAK 6

THE DISEASE- REACTIVE OPERATION



*You pay twice — once in treatment cost,
once in mortality and stunted growth you
cannot recover.*

LEAK NO. 06

The Disease-Reactive Operation

THE LEAK

You buy treatments when fish get sick. You don't run a prevention budget. So when an outbreak hits, you pay double — once in treatment cost, once in mortality and stunted growth.

Shrimp diseases are different. With fish, you see lesions, erratic swimming, gasping. With shrimp, the first sign of disease is often no sign at all. They just stop feeding. By the time you see white spots, red bodies, or black gills, you have already lost 20–40% of the batch.

The problem is not the disease. The problem is that you are not looking for pathogens before they arrive. You wait for symptoms. With shrimp, symptoms mean harvest early and lose money.

WHY IT'S HIDDEN

Treatment cost is visible (a bill). Prevention cost is invisible because it is not spent. Most operators feel virtuous about not spending on prevention, until an outbreak takes 30% of a batch.

PCR testing costs money. A full panel for WSSV, EHP, IMNV, *V. parahaemolyticus* might run \$200–500 per sample. Small operators skip it. They buy PLs from a hatchery with a piece of paper saying "PCR negative" and trust it. That paper is often weeks old. A lot can happen in two weeks.

The other hidden source: water. Even if your PLs are clean, your incoming water can carry pathogens. Wild crabs, birds, and even wind-blown sediment introduce viruses. You treat the pond, stock clean PLs, and still get an outbreak. You blame the hatchery. The hatchery was innocent. Your water killed you.

THE 30-MINUTE DIAGNOSTIC

Add up your last 12 months of treatment spending (chemicals, antibiotics, salt baths, emergency feed additives). If it is more than 1.5% of revenue, you are running reactive. If you have had any single outbreak that killed more than 10% of a batch, you are definitively reactive.

Pull your last three batch records. For each outbreak, answer: Did we PCR test PLs on arrival? Did we PCR test pond water before stocking? Did we quarantine PLs for any period before stocking?

If any answer is no, you are running reactive.

Then look at your treatment costs last 12 months. Divide by total harvest kg. If you are spending more than \$0.10 per kg on treatments (probiotics, antibiotics, disinfectants, lime, salt), you are treating problems you could have prevented.

THE FIX

Three preventive levers, in priority order:

1. Quarantine all incoming fingerlings for 7–14 days before stocking into production ponds.
2. Pond-side health checks every 3 days — gill, skin, behavior. Five minutes per pond.
3. Strategic immunostimulants (vitamin C, beta-glucan) at known stress points: stocking, transfer, weather changes.

Three other prevention steps specific to shrimp:

- 1- On-arrival PCR. Every batch. Random sample of 50 PLs, pooled into 5 tubes of 10. Test for WSSV, EHP, and Vibrio. Cost roughly \$300. If positive, reject the batch. That \$300 saves you \$30,000 in lost harvest.
- 2- Quarantine pond. One small pond or tank. Hold PLs for 5–7 days before stocking into production ponds. If they die in quarantine, they do not infect your whole farm.
- 3- Reservoir water treatment. If you pump water from a canal or well, hold it in a reservoir pond for 7–10 days before using it. Treat with chlorine or lime during that hold time. Let UV and settling work. Pathogens die without hosts. This one change cut disease incidence 60% on a farm I worked with in India.

REAL COST IMPACT

A single avoided 20% mortality event on a 5-tonne batch at \$2.20/kg = **\$2,200 saved**. Most operators experience 1–2 of these events per year. Prevention cost: typically \$300–600/year.

A farm with 4 cycles per year. Without prevention, one outbreak per year kills 25% of a batch. That batch would have been 15 tonnes. Loss = 3.75 tonnes at \$5/kg = \$18,750.

PCR testing for all four cycles at \$300 per cycle = \$1,200. Quarantine pond operating cost = roughly \$500/year. Total prevention = \$1,700.

That is a 10x return on a bad year. On a good year with no outbreak? You still sleep better.

My score for this leak: ____ / 3

LEAK 7 THE SOP GAP



*The leak that decides whether you ever scale
past \$50K/month — or stay trapped in your
own operation.*

LEAK NO. 07

The SOP Gap

THE LEAK

Every operation runs on the experience inside one or two people's heads. When that person is sick, on holiday, or quits, the operation degrades. New hires take six months to be useful instead of six weeks. Mistakes repeat across batches because nobody documented what worked last time.

Shrimp farming has more daily decisions than fish farming. Feeding rate changes with weather, molting cycle, DO, and appetite. Water exchange changes with tide (if you are tidal), salinity, and ammonia. Harvest timing changes with price, size, and survival curve.

All of these decisions live in the head of one person: the farm manager. When that person is off-site, decisions stop or become guesses. When that person leaves, you lose years of learning.

WHY IT'S HIDDEN

This isn't a daily cost. It's a compounding cost. It shows up as inconsistent results between batches, slow scaling, and operators who feel trapped in their own business.

Shrimp operators are often promoted from within. They learned by watching, not by reading. They know what to do but cannot explain why. When you ask them to write it down, they freeze. Not because they are hiding anything. Because they have never translated feel into procedure.

The cost shows up slowly. Inconsistent survival between ponds in the same batch. One pond does 12 tonnes, the next pond does 8 tonnes with the same stocking, same feed, same water. That difference is tribal knowledge that exists for one pond but not the other.

THE 30-MINUTE DIAGNOSTIC

Ask yourself: if you took two weeks off starting tomorrow, would the operation produce the same result? If the honest answer is no, you have an SOP gap. Walk the farm and count the number of decisions made each day that depend on tribal knowledge — feeding adjustments, water quality calls, harvest timing, fingerling assessment. Each one is a leak.

Pick a worker who has been on your farm less than 90 days. Ask them three questions:

1. "How do you decide when to increase feeding rate?"
2. "What do you check before starting a water exchange?"
3. "What do you do if shrimp stop feeding during a cloudy day?"

If they cannot answer each one with a clear step-by-step answer, your SOPs are missing or not being taught.

Then time yourself doing your morning pond check. Write down every decision you make. Every observation you note. Every threshold that triggers an action. That list is your hidden SOP. If it is not written down, it is not an SOP. It is just your memory.

THE FIX

Pick the five most repeated tasks. Document each one as a one-page SOP — bullets, not paragraphs. Photos help more than text. Test it: hand the SOP to your newest worker and watch them do the task. Update where they get stuck.

Create a **Daily Decision Log** – not a narrative, a checklist. For each pond, same time each day:

- DO (surface and bottom): ____
- Temperature: ____
- Salinity: ____
- Feed remaining on trays (yes/no): ____
- Shrimp behavior (normal/reduced feeding/at edge): ____
- Action taken: ____

That log becomes your SOP. After 30 days, the patterns emerge. You see exactly when the farm manager increases feeding (always after three consecutive days of clean trays). You see

when they exchange water (always when ammonia hits 0.5). You now have an SOP derived from behavior, not theory.

REAL COST IMPACT

SOPs are the leak that determines whether you ever scale past \$50K/month. Operations without SOPs cap themselves through pure operator burnout. With SOPs, the same operator typically grows revenue **30–60% in 12 months** without adding hours.

A farm with one experienced manager running 30 ponds. SOPs are in her head. She takes two weeks off for the first time in three years. In those two weeks, survival drops from 82% to 68% across the farm. Harvest is 7 tonnes lighter. At \$5/kg = \$35,000 loss.

Cost to document SOPs: roughly 20 hours of her time, plus a phone to record voice notes she transcribes later. Call it \$1,000 in her time.

That two-week vacation just cost 35 times the SOP documentation cost. And she still needed the vacation.

My score for this leak: ____ / 3

THE EXCUSE BUSTER

"I don't have time to measure all this."

You have time to lose money. Measuring FCR takes 10 minutes per batch if you keep a simple log. Measuring bottom DO takes 2 minutes per pond. You are not short on time. You are short on priority. If you cannot find 30 minutes a week to measure your biggest cost centers, you do not have a farm. You have an expensive hobby.

"My workers won't do the logs."

They will if you pay them for it. Or if you make it a condition of employment. Or if you do it yourself for two weeks to show them it takes 90 seconds per pond. The real problem is not your workers. The real problem is that you have not made measurement a non-negotiable. That is your job. Do it.

"I tried feed trays once and they didn't work."

What did you do when they did not work? Did you adjust feeding rate? Did you move the trays to a better location? Did you check them at the right time? A feed tray is not magic. It is a mirror. It shows you what you are doing wrong. If you looked in the mirror, did not like what you saw, and threw away the mirror, that is on you, not the tray.

"My farm is different. The benchmarks don't apply."

Every farm is different. And every farm still follows the same physics. Oxygen consumption per kg of shrimp is not negotiable. FCR math is not negotiable. Mortality

shows up on harvest weight whether you log it or not. Your farm is not special. It is either leaking or it is not. Run the numbers and find out.

"I'll fix it next cycle."

Next cycle becomes next cycle becomes next cycle. One year becomes five years. I have seen operators spend a decade saying "next cycle" while their margin drifted from 25% to 8% to 2% to red. The best time to fix a leak was last cycle. The second best time is today. Not next week. Not after harvest. Today.

"I don't have money for a DO sensor or PCR tests."

You have money for wasted feed and dead shrimp. You just do not see those as expenses because they do not arrive as invoices. A \$400 DO sensor saves you \$6,500 a year in electricity. That is a 16x return. If you cannot find \$400, your problem is not the sensor. Your problem is that you are already bleeding too much to see the ladder out of the hole. Fix a zero-cost leak first (SOPs, density recalc, feed trays). Use the savings to buy the sensor.

Your Scorecard

For each leak, score yourself 0–3.

0	Not a problem (I have data and the data shows we're fine)
1	Minor (suspected drift, no major impact yet)
2	Active leak (visible cost or symptom)
3	Bleeding (urgent, costing material money)

#	Leak	Score (0–3)
1	Feed Conversion Drift	_____
2	Invisible Mortality	_____
3	Stocking Density Misfit	_____
4	Water Exchange Black Hole	_____
5	Energy: Silent Margin Killer	_____
6	Disease-Reactive Operation	_____
7	SOP Gap	_____
TOTAL		____ / 21

READING YOUR SCORE

0–4	You're running a tight operation. Focus on the one or two scoring above 0.
5–10	Meaningful margin to recover. Pick the highest-scoring leak and fix it this quarter.
11–15	Your operation is leaking heavily. The fixes here can typically recover 15–30% of margin within 90 days.
16+	Stop scaling. Fix leaks before adding any new capacity.

THE LEAK MAP: Where To Pull First

Think of your operation as a pipe. Leaks upstream make downstream leaks look worse than they are. Fix upstream first, or you will keep chasing symptoms.

UPSTREAM LEAKS (Fix These First)

- **Leak 3 (Stocking Density Misfit)** – If your density is wrong, everything downstream is wrong. FCR, mortality, water exchange, energy. All of them get worse because you have either too many shrimp struggling or too few shrimp covering fixed costs.
- **Leak 6 (Disease-Reactive)** – Disease eats margin at the source. It raises mortality (Leak 2), increases FCR because sick shrimp do not convert feed (Leak 1), and often triggers unnecessary water exchanges (Leak 4) from panic.

MIDSTREAM LEAKS (Fix Second)

- **Leak 1 (Feed Conversion Drift)** – Fixing FCR reduces organic load, which reduces ammonia, which reduces the need for water exchange (Leak 4) and aeration (Leak 5).
- **Leak 2 (Invisible Mortality)** – Fixing mortality gives you more biomass for the same feed, which improves your FCR calculation even if feeding stays the same.

DOWNSTREAM LEAKS (Fix Last, Or They Reappear)

- **Leak 4 (Water Exchange)** – If you fix exchange but do not fix density or FCR, your ammonia will still spike and you will go back to old habits.
- **Leak 5 (Energy)** – If you install timers and DO sensors but are overstocked (Leak 3), you will still need full aeration. The savings will not appear.
- **Leak 7 (SOP Gap)** – SOPs document the way you run the farm after it is fixed. Writing SOPs before fixing leaks just documents broken processes.

THE SHORTCUT

If you have only 90 days and want the biggest return, fix in this order:

1. Leak 3 (Density) – Quick calculation, zero cost to diagnose.
2. Leak 1 (FCR) – Feed trays and weekly sampling, under \$100.
3. Leak 6 (Prevention) – PCR and quarantine, under \$500.

Those three alone typically recover 15–25% margin within one cycle. Then use that margin to fund the rest.

What To Do Next - THREE DOORS, ONE PATH

You finished the audit. You have your score. Now what?

Three options, in order of cost.

OPTION 1

Fix it yourself. (free).

Take the highest-scoring leak. Apply the diagnostic. Implement the fix. Measure for 90 days. Most operators recover 5–15% margin this way. No money spent. Just discipline.

OPTION 2

The 60-Minute Diagnostic Call – \$\$

You have the scorecard. But you want to know exactly where to swing the hammer first. And you want to talk to someone who has seen your exact problem before.

On this call, I will:

- Review your scorecard and your last three batch numbers
- Identify the single leak with the highest ROI to fix first
- Give you a specific action plan for the next 30 days
- Tell you honestly if you need Door 3 or not

If after 30 minutes you think I am full of it, we end the call and I refund your \$500. No hard feelings.

Book the call here: [Book Strategy 60 minute session](#)

OPTION 3**Bring me In : The 90-Day Farm Transformation Program.**

This is not coaching. This is hands-on restructuring.

I work with a small number of aquaculture operators each year who are doing \$30K+/month and scoring above 11 on the audit. We do not talk about mindset. We do not talk about goals. We fix the leaks, in order, on your farm.

Deliverables include:

- Full operational audit (on-site or remote with video)
- Fixed FCR and mortality tracking systems installed
- SOP library written for your farm, not a template
- Stocking density and aeration recalculations pond by pond
- Weekly review calls for 90 days
- An investor-ready operations manual

Investment: Starts at \$15,000 for the 90 days. Paid in three milestones.

If you are scoring 16+ and scaling is killing you, this is the door.

Apply here: [Book a consultation for farm Improvement](#)

(I only take three operators per quarter.)

Which door should you pick?

- Score 0–4: Door 1. You are fine. Just tune the one leak above zero.
- Score 5–10: Door 1 or Door 2. The \$500 call will save you months of trial and error.
- Score 11–15: Door 2 first. Then Door 3 if the call confirms you need deep restructuring.
- Score 16+: Door 3. Do not pass go. Do not fix it yourself. You are bleeding too fast to learn on the job.

***Whatever you choose, do one thing:
fix the highest-scoring leak before the end of this quarter.
Compounded over a year, that single fix will outperform
every '10x growth strategy' course on the market.***

Hazem

About the Author

Hazem is a Director and Business Growth Architect with 30+ years of hands-on, globally-deployed expertise across aquaculture, regenerative agriculture, and sustainable food systems.

He has restructured under-performing agri-ventures across three continents, taking small operators from \$15K/month plateaus into investment-ready, \$100K+/month operations.

His work focuses on the unsexy levers — margin, mortality, FCR, SOPs, capacity — that move a real operation. He doesn't sell mindset. He sells systems that work in the water and on the ground.

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THE WALL SHEET:

7 Leaks, 7 Fixes, 7 Weekly Checks

Leak	The 30-Minute Fix (Do It Once)	The Weekly Check (5 minutes)
1. Feed Conversion Drift	Install feed trays in largest pond. Reduce feeding if trays have leftover pellets.	Pull trays at same time daily. Log "clean" or "dirty".
2. Invisible Mortality	Start daily visible mortality log. Seine one pond bottom weekly.	Compare daily log to harvest survival. Gap = invisible loss.
3. Stocking Density Misfit	Recalculate carrying capacity using real aerator output, not rated. Add 30% safety margin.	Measure bottom DO at 4am weekly. If below 4 mg/L, density too high.
4. Water Exchange Black Hole	Switch from fixed exchange to trigger-based (ammonia >1.0 or nitrite >0.5).	Test ammonia and nitrite at peak feed time weekly.
5. Energy Killer	Install one DO sensor with relay on largest pond. Set aerators to cycle off above 6 mg/L.	Calculate kWh per kg harvested weekly. Compare to benchmark.
6. Disease-Reactive	PCR test PLs on arrival. Quarantine 7 days before stocking.	Walk pond edges daily. Look for stopped feeding or edge shrimp.
7. SOP Gap	Document five most repeated tasks as one-page bulleted checklists.	Hand one SOP to a worker. Watch them do it. Update where stuck.

YOUR WEEKLY RHYTHM (Total: 30 minutes)

- Monday: Run the weekly check for your highest-scoring leak.
- Wednesday: Run the weekly check for your second-highest leak.
- Friday: Run the weekly check for your third-highest leak.
- Saturday: Review your log. Did any number cross a threshold? Act.

IF YOU DO NOTHING ELSE, DO THIS

Pick one leak from the scorecard. Do the 30-minute fix this week. Do the weekly check every week for 90 days. Measure the difference. Then pick the next leak.

That sequence, repeated seven times, will change your business more than any course, any consultant, any new equipment. Because it is not about the leaks. It is about building the habit of seeing them before they bleed you dry.

*****Disclaimer*****

This book is based on field observations, operational data, and lessons drawn from real aquaculture projects across multiple regions and production systems. The examples, numbers, and scenarios used throughout are either composites or have been anonymized to protect the operators involved.

Nothing here is a guaranteed formula. Shrimp don't read manuals, and every farm — its water, its feed, its labor, its location — is different. What reversed a profit bleed on one farm may not transfer directly to yours without adjustment.

This is not legal, financial, or veterinary advice. If you're making a major operational or investment decision, bring in a qualified professional who can assess your specific conditions on the ground.

The goal of this book is to help you ask better questions and see problems you might be walking past every day. What you do with that is on you.

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